

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7				Indicative content - ions are free to move because aluminium oxide is molten - ions travel to the oppositely charged electrodes because opposite charges attract / Al^{3+} ions go to the cathode / O^{2-} ions go to the anode - reduction is gain of electrons, oxidation is loss of electrons at the cathode – reduction $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ - aluminium ions gain 3 electrons to form aluminium atoms at the anode – oxidation $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ - two oxide ions lose 2 electrons each to form two oxygen atoms - these pair up to form oxygen molecules 5-6 marks Good explanation of the movement of ions towards both electrodes and of reduction and oxidation of both ions in terms of gain/loss of electrons; attempt at electrode equations <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		

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					AO1	AO2	AO3	Total	Maths	Prac
				3-4 marks Attempt at explanation of the movement of ions and of reduction and oxidation in terms of gain/loss of electrons <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Simple description of the movement of ions during the process; knowledge of reduction or oxidation in terms of gain/loss of electrons <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 7 total	6	0	0	6	0	0